PULLBACK OF K-FLAT COMPLEXES WHEN THE SITE HAS ENOUGH POINTS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument; the reduction and the pointwise computation are both inside the counted span.

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Editorial note. The checked chapter list for same-name collisions is the set of stacks chapters staged in this repository: chow, cohomology, divisors, etale-cohomology, flat, more-algebra, more-morphisms, morphisms, spaces-more-morphisms, varieties together with this card's chapter sites-cohomology.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

Allowed standard prerequisites (editorial).

- Enough points for a site and the interpretation of exactness on stalks. Same chapter together with the chapter Sites, the notion of a site with enough points, for which acyclicity can be tested on stalks. Supplies the hypothesis under which K-flatness is checked pointwise.
- K-flat complex and the flatness criterion by stalks. Same chapter, the definition of K-flatness and the companion criterion characterising it by the behaviour of the stalks when the site has enough points. Supplies the criterion applied after pullback.

Source reference. Original lemma 0DEN: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 3767–3767. Navigation pointer invoked at proof line 3842; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 05V5: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 3843–3843. Navigation pointer invoked at proof line 3843; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

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Source reference. Original lemma 06Y1: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 3845–3845. Navigation pointer invoked at proof line 3845; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Lemma 0.1. *Let $f : (Sh(\mathcal{C}), \mathcal{O}) \to (Sh(\mathcal{C}'), \mathcal{O}')$ be a morphism of ringed topoi. If $\mathcal{C}$ has enough points, then the pullback of a K-flat complex of $\mathcal{O}'$-modules is a K-flat complex of $\mathcal{O}$-modules.*

Proof. This follows from Lemma 0DEN, Modules on Sites, Lemma 05V5, and More on Algebra, Lemma 06Y1. □